

A PROJECT TITLE

ON

Traffic Flow Pattern Clustering Using K-Means and DBSCAN for Smart Urban Traffic Congestion Analysis

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

N. Yeswanth Kumar 2520030183

B. Sai Sashank 2520030454

Under the Esteemed guidance of

Mr. Shaik Asif

Department of

CSE



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus, Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

Name of Title: Traffic Flow Pattern Clustering Using K-Means and DBSCAN for Smart Urban Traffic Congestion Analysis

Abstract:

Traffic congestion has become one of the major challenges in rapidly growing urban areas. Increasing numbers of vehicles on roads result in traffic delays, fuel wastage, environmental pollution, and reduced productivity. Traditional traffic monitoring systems often struggle to analyze large volumes of traffic data efficiently and identify meaningful traffic patterns. Machine Learning provides intelligent techniques to analyze traffic data and support effective traffic management. This project proposes a traffic flow analysis system using two popular unsupervised machine learning algorithms, **K-Means** and **DBSCAN**. K-Means clusters traffic data into groups based on similar characteristics such as vehicle count, average speed, traffic density, and road occupancy. It helps identify recurring traffic patterns during different periods of the day. DBSCAN, a density-based clustering algorithm, detects irregular traffic congestion areas and identifies noise or unusual traffic events that may occur because of accidents, roadblocks, or special events. The traffic dataset is first preprocessed by handling missing values, removing duplicates, and normalizing numerical attributes. Feature selection is performed to improve clustering accuracy. K-Means is applied to divide the traffic flow into predefined clusters, while DBSCAN discovers naturally occurring congestion regions without requiring the number of clusters beforehand. The performance of both algorithms is evaluated using clustering quality measures such as Silhouette Score, Davies-Bouldin Index, and visual cluster representation. The proposed system enables traffic authorities to understand traffic behavior, identify congestion hotspots, and support data-driven traffic management decisions. The clustering results can assist in optimizing traffic signals, planning alternate routes, and improving road infrastructure. Overall, this project demonstrates how unsupervised machine learning techniques can effectively analyze complex traffic datasets and contribute to the development of intelligent transportation systems for smart cities.

1. INTRODUCTION:

Traffic congestion is one of the most significant transportation problems faced by modern cities. Rapid urbanization and the continuous increase in vehicle ownership have led to severe traffic jams, longer travel times, increased fuel consumption, and higher air pollution. Efficient traffic management has therefore become an essential requirement for smart city development.

Machine Learning techniques enable computers to analyze traffic data without requiring explicit programming. Among these techniques, clustering algorithms are widely used to discover hidden patterns in unlabeled datasets. This project focuses on two clustering algorithms: K-Means and DBSCAN.

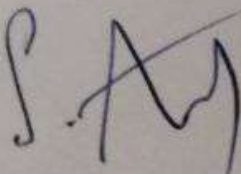
K-Means partitions traffic data into a fixed number of clusters based on similarity, making it useful for identifying regular traffic flow patterns. DBSCAN groups data points according to density and can identify congestion hotspots and unusual traffic events while treating isolated points as noise.

The proposed system collects traffic-related attributes such as vehicle count, speed, traffic density, and occupancy. After preprocessing and feature normalization, both clustering algorithms are applied to analyze traffic conditions. The clustering results help identify low, medium, and high traffic zones and detect abnormal congestion.

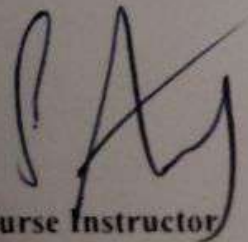
The project aims to support intelligent traffic management by providing valuable insights into traffic behavior. The generated clusters can help transportation authorities optimize traffic signal timings, recommend alternate routes, and improve overall road planning for safer and more efficient transportation.

2. SOFTWARE REQUIREMENTS:

Component	Requirement
Operating System	Windows 10/11 or Ubuntu Linux
Programming Language	Python 3.10 or above
IDE	Jupyter Notebook / Visual Studio Code / PyCharm
Machine Learning Library	Scikit-learn
Data Analysis	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Dataset Source	Kaggle / UCI Machine Learning Repository / Traffic Sensor Data
Database (Optional)	MySQL
Version Control	Git & GitHub
Hardware	Intel Core i3/i5 or above, 8 GB RAM, 10 GB free storage



Signature of the Guide



Course Instructor